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SARDAR PATEL UNIVERSITY

BCA (Semester-V) On-Demand Examination – February 2026

[US05CBCA51]: Object Oriented Programming - III

Date: 9/2/2026 (Monday)

Time: 10.00am to 1.00pm

Total Marks: 70

Note: Figure indicate at right is maximum marks for question.

Q -1 Select an appropriate option for the following statements.

[10]

1. Which method of DriverManager class is use to open a connection.
[a] openConnection() [b] setConnection()
[c] makeConnection() [d] getConnection()
2. The JDBC _____ class defines objects which can connect Java applications to a JDBC driver.
[a] Statement [b] DriverManager
[c] ResultSet [d] SQLManager
3. The indexOf() method returns the _____ if element is not present in the list.
[a] 0 [b] True
[c] False [d] -1
4. The life cycle of a servlet is managed by _____.
[a] Servlet Context [b] Servlet Container
[c] The Supporting Protocol [d] All of these
5. What is javax.servlet.Servlet?
[a] Concreate class [b] Abstract class
[c] Interface [d] None of these
6. Which of the following code is used to get cookie in servlet?
[a] response.getCookies() [b] request.getCookies()
[c] Cookies.getCookies() [d] None of these
7. The difference between Servlet and JSP is the _____.
[a] Translation [b] Compilation
[c] Syntax [d] Both [a] & [b]
8. TLD Stands For _____.
[a] Tag Library Data [b] Tag Library Domain
[c] Tag Library Descriptor [d] Tag Library Description
9. The life cycle of thread in java is controlled by _____.
[a] JVM [b] JDK
[c] JRE [d] None of these
10. What is sometime called a lightweight process?
[a] Process [b] Thread
[c] JVM [d] All of these

[4]

Q-2 Answer the following questions in brief. (Attempt any Ten)

[20]

1. Explain JDBC.
2. List types of Drivers, available in JDBC API.
3. Explain ResultSet with its type.
4. Explain init().
5. Draw web-based architecture.
6. Briefly explain getServletInfo() method of Servlet.
7. What is JSP? List advantages of JSP.
8. What is JSP Expression?
9. Draw the structure of JSP Life Cycle.
10. **Define:** Thread scheduling.
11. List Java Thread methods.
12. Explain void setPriority().

Q-3

[A] Draw and explain JDBC architecture in detail.

[05]

[B] Write a step of JDBC program for executing any query?

[05]

OR

[A] Explain Statement object and ResultSet object.

[05]

[B] Explain ArrayList in detail with example.

[05]

Q-4 Explain Servlet Life Cycle through diagram.

[10]

OR

Q-4 Explain Servlet request and response method in detail.

[10]

Q-5

[A] Explain the execution process of JSP in detail.

[05]

[B] Compare JSP with Servlet.

[05]

OR

Q-5

[A] Write a note on JSP components.

[05]

[B] Discuss JSP page Directives with all its attributes.

[05]

Q-6

[A] Draw and discuss Thread Life Cycle.

[05]

[B] What is Thread? Discuss use of thread with its characteristics.

[05]

OR

[A] Write a note on Java Thread implementation.

[05]

[B] What is Thread Synchronization? Explain how to implement synchronization using example. **[05]**

[2]